

# Prem Mallappa

Systems Software Architect • Embedded Systems • Virtualization

Bengaluru, India

+91-9448900326

[prem.mallappa@gmail.com](mailto:prem.mallappa@gmail.com)

[pmallappa.github.io](https://pmallappa.github.io)

[pmallappa](#)

[pmallappa](#)

Seasoned Software Architect with extensive expertise in designing and developing high-performance embedded systems, system software, Linux kernel drivers, and virtualization technologies. Proficient in a wide range of programming and scripting languages, with a proven track record of delivering robust, scalable solutions in complex technical environments. A highly analytical and collaborative team player, recognized for strong problem-solving abilities, logical thinking, and a passion for mastering emerging technologies. Committed to driving innovation and excellence through clean architecture, optimized code, and best-in-class engineering practices.

## Skills

|                                       |                                                                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Leadership</b>                     | <b>22+ years</b> of multidisciplinary team and project leadership experience; <b>software architecture</b> and technical direction across systems and platform teams.                  |
| <b>Communications</b>                 | <b>Clear written and spoken communication</b> for architecture reviews, design discussions, mentoring, and cross-functional execution.                                                 |
| <b>Technology</b>                     | Broad and deep IT expertise across <b>embedded systems, system software, cloud/security, virtualization</b> , and software services development.                                       |
| <b>Computer Security</b>              | <b>Cryptography</b> , virtualization and cloud computing security architecture, <b>secure software development</b> , risk management, compliance, intrusion detection, and prevention. |
| <b>Programming Languages</b>          | <b>C, C++</b> , Assembly ( <b>x86, ARM, MIPS</b> ), Rust, Python, Go, BASH                                                                                                             |
| <b>Operating Systems</b>              | <b>Linux</b> , FreeBSD, Symbian                                                                                                                                                        |
| <b>RTOS</b>                           | <b>VxWorks, QNX</b> , seL4, Zephyr, ThreadX                                                                                                                                            |
| <b>Architecture</b>                   | <b>x86, ARM v6/v7/v8, MIPS64</b> , PowerPC, RISC-V                                                                                                                                     |
| <b>Development Tools</b>              | <b>GCC, Clang, Git, CMake</b> , Make, GDB, Perf, VTune, ARM Development Tools, Keil, JTAG, Docker, QEMU                                                                                |
| <b>Systems &amp; Virtualization</b>   | <b>Linux kernel development, device drivers</b> , hypervisors, <b>IOMMU/SMMU</b> , pre-silicon validation, boot flows, and low-level firmware bring-up.                                |
| <b>Performance &amp; Acceleration</b> | <b>SIMD/vectorization, performance optimization</b> , math and crypto libraries, <b>SmartNIC/DPU offload</b> , NVMe-oF, RDMA, and hardware-assisted data paths.                        |
| <b>Specialized Skills</b>             | Embedded systems, system software, <b>cryptography</b> , filesystems, performance engineering, packet/data-path acceleration, and platform enablement.                                 |

## Education

### Masters (Computer Science)

BITS Pilani

Pilani, Rajasthan, India

2016

### Bachelors (Computer Science)

Adichunchangiri Institute of Technology (AIT)

Affiliated: Visvesvaraya Technological University (VTU)

Chikkmagalur, Karnataka, India

2002

# Experience

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## AMD Ltd.

2018 – Present

Bengaluru, India

### PRINCIPAL ENGINEER

Worked in multiple teams within AMD including **AMD LibM**, **AOCL Cryptography**, **BSP**, and **NVMe** in **Pensando** BU. Every project was different, requiring quick learning and adaptation to new domains. The role also involved **leading a team** of four-to-six engineers in various projects. I was also part of the leadership team where I mentored engineers from across AOCL and Pensando BU.

### SR. MEMBER OF TECHNICAL STAFF

Worked on **Model0** which was **pre-silicon validation** of next generation AMD cores, including Zen2 and Zen3. The role involved working with hardware team to bring up bare-minimum processor configuration with just core and memory controller, and run Linux, stressors and benchmarks. This involved working on low-level firmware, bootloaders, Linux kernel and performance analysis tools.

## Broadcom Ltd.

2014 – 2016

Bengaluru, India

### PRINCIPAL ENGINEER

Architected foundational software stack for Broadcom Vulcan, a multicore-multithreaded ARMv8 64-bit server processor. Engineered complete **SMMUv3** (System Memory Management Unit) driver infrastructure supporting advanced IOMMU virtualization. Developed and contributed an **SMMUv3 emulation model** to QEMU mainline, enabling early platform validation and ecosystem enablement. Implemented comprehensive virtualization features including command queue processing, STE/CD descriptor parsing, multi-level page table walks, and stage-1/stage-2 address translation.

## Cavium India Pvt. Ltd.

2011 – 2013

Bengaluru, India

### TECH LEAD

Led low-level software development for Cavium Octeon III (MIPS64) multicore network processors, delivering stable platform firmware for networking customers. Resolved a critical KEXEC kernel bug and authored the full **MIPS64 port** of the Kexec/Kdump crash dump infrastructure, ultimately merged into Linux mainline. Engineered a bare-metal core dump generation framework with a host-resident daemon that enabled detailed post-mortem GDB analysis during bring-up. Designed and implemented the **CavHv hypervisor** for MIPS64 architecture using hardware virtualization extensions to validate next-generation features.

## ARM Ltd.

2005 – 2009

Bengaluru, India

### SR. DEVELOPMENT ENGINEER

Executed comprehensive **OS porting initiatives** for emerging ARM architectures including ARM1176JZFS with TrustZone security extensions and the Cortex-A8 application processor. Developed a production-grade touchscreen driver for Symbian OS along with a Python-based automated validation framework used across multiple programs. Designed a precision interrupt latency measurement infrastructure that quantified TrustZone secure-world context switching overhead for silicon partners. Ported the **L4Ka::Pistachio microkernel** to ARM architecture, enabling virtualization research and multiple proof-of-concept implementations.

## Sasken Communication Technologies Ltd.

2004 – 2005

Bengaluru, India

### SR. SOFTWARE ENGINEER

Architected and maintained the **Extended File System (EFS)** for VxWorks-based UMTS base station infrastructure, balancing throughput with resiliency requirements. Engineered a reset-resilient filesystem with **advanced wear-leveling algorithms** so data integrity held across unexpected power cycles in the field. Designed a flash-optimized flat file tree structure that minimized write amplification and significantly extended device longevity.

## Global Edge Software Ltd.

2003 – 2004

Bengaluru, India

### SOFTWARE ENGINEER

Designed and implemented a **comprehensive SDIO host controller driver** for Linux on Intel StrongARM embedded platforms used in consumer devices. Engineered a high-performance 4-bit SDIO driver for Marvell 802.11g WiFi chipsets that consistently achieved maximum theoretical throughput. Optimized **DMA transfers** and interrupt handling paths to minimize latency and maximize wireless data transfer efficiency under heavy load.

## Short Stints

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### VSPL Ltd.

Software Architect

Nov. 2016 – Jan. 2018

Bengaluru, India

### Cisco Ltd.

Software Engineer

Oct. 2010 – Apr. 2011

Bengaluru, India

### B-Labs, London UK

Sr. Engineer - Contractor

Nov. 2009 – Sep. 2010

Bengaluru, India

### Harman International

Engineer

Jul. 2009 – Nov. 2009

Bengaluru, India

## Open Source

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### QEMU [\(Link\)](#)

2014-2016

- SMMUv3 (IOMMU) emulation support for ARMv8
- Designed and implemented SMMUv3 model merged into mainline
- Added support for Stage1, Stage2, and nested virtualization
- Implemented command queue processing and page table walk

### Linux Kernel (IOMMU/SMMUv3)

2015-2016

Contributed to upstream SMMUv3 driver for Broadcom Vulcan

- [Fix GERRORN register write](#)
- [Fix STE.VALID check in Vulcan](#)
- [Workaround for STE.VALID](#)

### Linux Kernel (Kexec/Kdump) [\(Link\)](#)

2011-2013

- MIPS Kexec/Kdump port for Cavium Octeon platforms
- Developed MIPS64 port of Kexec and Kdump (merged upstream)
- Fixed critical bugs in KEXEC for Cavium Octeon platforms

### GLIBC [\(Link\)](#)

2018-2020

- Performance optimizations for AMD processors
- Fixed memcpy behaviour on AMD processors
- Optimized string functions for x86-64 architecture
- Performance improvements for memory operations

### AMD LibM [\(Link\)](#)

2018-2025

- Open source math library for AMD processors
- Core contributor to open sourcing AMD Math Library
- Optimized transcendental functions (exp, log, pow, trig)
- Implemented SIMD/FMA optimizations for vector operations

# Projects

## AMD Ltd.

2018 – Present

### PRINCIPAL ENGINEER

Bengaluru, India

#### AMD Pensando DSC Live Migration

Jan. 2022 – Present

**Synopsis** Delivered fully supported live-migration feature for the AMD Pensando DSC SmartNIC across all host-visible device modes — network, virtio, and NVMe. Work spanned Linux kernel control-path frameworks, P4+-based dataplane with hardware dirty-page tracking, and scalability validation on Elba ASICs to achieve zero-downtime VM migration in cloud environments.

**Details** Engineered vendor control-path support across the **Linux VFIO and vDPA frameworks**, ensuring correct migration sequencing for all device presentation modes. Implemented the migration datapath entirely in **P4+ language** with **hardware-level dirty-page tracking** to minimize pre-copy transfer volume and downtime. For the vDPA path, integrated **system IOMMU dirty-tracking** as the page-change source in place of device hardware, fully closing the feature gap. Validated both VFIO and vDPA combinations end-to-end using **QEMU-orchestrated migration across multiple physical hosts**. Subsequently investigated **scalability bottlenecks** on AMD/Pensando Elba ASICs by tracing dirty-page reporting through the **SEQ/ACK bitmap protocol** in the dataplane, firmware, and host migration drivers. Built an **end-to-end validation framework** for NVMe + virtio-pci VM topologies with automated QEMU launch, VF bind checks, hugepage preparation, and **QMP-driven pre-copy / stop-and-copy orchestration**.

**Languages:** C, C++, P4, Python  
**Tools:** Git, GDB, Qemu, Xen Hypervisor, DPDK, Linux Kernel  
**Tech:** VFIO, VDMA, IOMMU, NVMe-oF, SmartNIC, DPU, P4+, virtio, Packet Processing Pipelines

#### AMD Pensando DSC Platform PCIe support

Jun. 2025 – Dec. 2025

**Synopsis** The DPU presents itself as a PCIe device, all access from the host ends up in indirect access and handled in software stack. Optimizing the PMT/PRT entries to make most of the hardware access end up as direct memory access.

**Details** Analyzed the host-to-DPU PCIe access path and identified where **indirect register and memory accesses** were introducing avoidable software overhead. Reworked **PMT/PRT programming** so frequently used host accesses could be resolved as **direct memory transactions** instead of trapping into slower software-managed flows. Validated the updated PCIe mapping behavior across host driver interactions and DPU-side handling to reduce access latency, simplify the datapath, and improve overall platform responsiveness for control-plane and bring-up operations.

**Languages:** C, C++, P4, Python  
**Tools:** Git, GDB, Qemu, Xen Hypervisor  
**Tech:** PCIe, SmartNIC, DPU, Hardware Offload

#### AMD Cryptography Library

Jan. 2023 – Jun. 2025

**Synopsis** Initiated as proof-of-concept to address ISV performance bottlenecks with AES-CFB cryptography on AMD EPYC Milan servers. Evolved into comprehensive cryptographic primitives library delivering production-grade implementations of AES and SHA algorithms optimized for AMD processor architectures.

**Details** Engineered parallel AES-CFB decryption algorithm delivering **3x performance improvement** over standard OpenSSL implementation. Developed multi-algorithm cryptographic library supporting AVX2, AESNI, and SHANI instruction set extensions. Implemented **CPUID-based runtime dispatcher** for automatic selection of optimal algorithm variant per processor generation. Integrated library as OpenSSL provider enabling transparent drop-in replacement for existing applications.

**Languages:** C++, x86-64 Assembly  
**Tools:** Git, CMake, GCC, Clang, OpenSSL Provider API  
**Tech:** AES-CFB, AES-GCM, SHA256, SHA512, AVX2, AVX-512, AESNI, SHANI, VAES

#### AMD Math Library (AOCL-libm)

Dec. 2019 – Dec. 2022

**Synopsis** High-performance mathematical library providing optimized implementations of transcendental functions (exp, log, pow, trigonometric) for AMD EPYC and Ryzen processors. Led comprehensive modernization initiative to enhance performance through advanced algorithmic techniques and microarchitecture-aware optimizations.

**Details** Implemented **Estrin's polynomial evaluation** method enabling parallel FMA instruction utilization for reduced latency. Developed vectorized lookup table architecture with cache-line alignment and interleaving for optimal memory bandwidth. Achieved **30% performance improvement** in scalar operations and **2x improvement** in vector operations across function portfolio. Engineered CPUID-based dynamic dispatcher selecting optimal code paths based on processor generation and feature flags.

**Languages:** C, x86-64 Assembly, Python  
**Tools:** CMake, GCC, Clang, AOCC, Git, Perf, VTune  
**Tech:** AVX2, AVX-512, FMA3, FMA4, Estrin Polynomial, SIMD Vectorization, Cache Optimization

**Model0**

Feb. 2018 – Dec. 2019

- Synopsis** Next generation AMD processors needed to be tested before IO and other components are integrated, as part of the **shift-left** strategy. Allows faster integration and
- Details** As part of the Shift-Left strategy, full OS boot and CPU stressers were run early in the development before tape-out, with just the compute core and simulated memory, but no I/O. stressing and lmbench etc were run to stress the CPU code path. Efforts involved trimming down Linux kernel enough so that it boots **without ACPI**, and in some cases with a very small ACPI table.

**Languages:** C, x86-64 Assembly, Python  
**Tools:** CMake, GCC, Linux Kernel, Virtualization  
**Tech:** AMD-SEV, AMD-SNP

**Broadcom Ltd.**

2014 – 2016

**PRINCIPAL ENGINEER**

Bengaluru, India

**ARM IO Virtualization with SMMUv3**

Jan. 2015 – Oct. 2016

- Synopsis** Developed comprehensive SMMUv3 (ARM System Memory Management Unit) emulation infrastructure enabling early software development and validation for Broadcom Vulcan ARMv8 server platform. SMMUv3 provides hardware-based IO virtualization with multi-stage address translation and device isolation capabilities.
- Details** Architected and implemented complete **SMMUv3 device model** for QEMU, successfully **upstreamed to mainline** repository. Developed command queue processing engine supporting asynchronous invalidation and event notification. Implemented ARMv8 LPAE page table walker supporting Stage-1, Stage-2, and nested translation modes. Designed comprehensive event reporting mechanism with configurable interrupt generation and fault handling.

**Languages:** C  
**Tools:** QEMU Device Model, Git, GDB, ARM Fast Models  
**Tech:** ARMv8, SMMUv3, IOMMU, 2-Stage Translation, Command Queue, Event Records

**SMMUv3 Driver for Broadcom Vulcan**

Jan. 2015 – Oct. 2016

- Synopsis** Broadcom Vulcan represents ground-up ARMv8 server processor design featuring 32 cores with simultaneous multithreading (4 threads per core). Engineered production Linux driver for integrated SMMUv3 enabling secure IO virtualization and DMA protection for datacenter workloads.
- Details** Developed initial **Linux kernel driver** implementing SMMUv3 core functionality including StreamID-based device isolation. Collaborated cross-functionally with hardware design team to identify and resolve silicon errata and performance bottlenecks. Contributed bug fixes and feature enhancements to ARM's reference SMMUv3 driver implementation. Designed and executed a **comprehensive validation suite** covering multi-stage translation and fault handling scenarios.

**Languages:** C  
**Tools:** Git, GCC, ARM Cross Compiler, Kernel Build System  
**Tech:** Linux Kernel Driver, IOMMU Framework, DMA API, Device Tree, MSI Interrupts

**TECH LEAD**

Bengaluru, India

**Broadcom XLP (MIPS64) Platform Enablement**

Jan. 2014 – Dec. 2015

- Synopsis** Broadcom XLP represents high-performance multicore MIPS64 processor architecture featuring up to 32 cores with 2-way multithreading and integrated hardware acceleration engines for networking and security workloads. Led Linux kernel enablement delivering production-grade device drivers and power management infrastructure.
- Details** Developed hardware random number generator driver providing high-entropy source for cryptographic applications. Implemented **Common Clock Framework** driver enabling dynamic frequency scaling across processor domains. Engineered DVFS (Dynamic Voltage and Frequency Scaling) driver reducing **power consumption by 40%** under variable workloads. Created CDE (Compression/Decompression Engine) driver offloading network packet processing to hardware accelerators.

**Languages:** C, MIPS Assembly  
**Tools:** Git, GCC, MIPS Cross Compiler, Device Tree Compiler  
**Tech:** MIPS64 Octeon, Common Clock Framework, DVFS, Hardware RNG, CDE Engine

## Cavium Networks

2011 – 2013

### TECH LEAD

Bengaluru, India

#### Cavium OCTEON III - Kexec/Kdump Implementation

May, 2011 – Dec, 2013

- Synopsis** Kexec enables fast kernel reboot without firmware re-initialization, while Kdump leverages this mechanism for capturing crash dumps. Developed complete MIPS64 architecture port enabling zero-downtime kernel updates and comprehensive crash analysis for Cavium OCTEON network processors.
- Details** Architected and implemented complete **MIPS64 port** of Kexec/Kdump infrastructure supporting OCTEON's Hybrid SMP topology. Resolved critical architectural bug preventing Kexec functionality on MIPS platforms. Successfully **upstreamed two kernel patches** to Linux mainline, now universally deployed across MIPS64 systems. Enabled seamless kernel hot-patching without disrupting Hybrid SMP core configurations or network traffic.

**Languages:** C, MIPS64 Assembly  
**Tools:** Git, GDB, Crash Analysis, Kernel Build System  
**Tech:** Kexec, Kdump, ELF Boot Protocol, Hybrid SMP, Memory Reservation, Crash Kernel

#### CavHv Hypervisor for MIPS64

Jan, 2012 – Dec, 2013

- Synopsis** Experimental Type-1 hypervisor leveraging MIPS Virtualization (VZ) hardware extensions for Cavium OCTEON-III processors. Collaborated with MIPS architecture team and Cavium hardware designers to validate draft virtualization specification through practical implementation.
- Details** Designed and implemented **bare-metal Type-1 hypervisor** based on MIPS Virtualization Extensions draft specification. Developed guest context management supporting CP0 register virtualization and root/guest mode transitions. Successfully demonstrated **concurrent execution of two independent Linux kernel instances** with isolated address spaces. Provided critical architectural feedback influencing finalization of MIPS VZ specification.

**Languages:** C, MIPS64 Assembly  
**Tools:** GCC, MIPS Toolchain, GDB, QEMU  
**Tech:** MIPS VZ Extensions, Type-1 Hypervisor, CP0 Virtualization, Root/Guest Mode

## Cisco Systems

Oct, 2010 – Apr, 2011

### SOFTWARE ENGINEER

Bengaluru, India

#### Memory Leak Detection for Cisco IOS

Oct, 2010 – Apr, 2011

- Synopsis** Cisco IOS operates in both bare-metal and Linux environments with multi-month uptimes, making memory leak detection critical. Developed non-intrusive garbage detection system enabling proactive identification of unreachable memory in long-running network infrastructure.
- Details** Architected **custom C runtime wrapper** intercepting malloc/free operations without application source modifications. Implemented concurrent monitoring thread executing intrusive mark-and-sweep algorithm during idle periods. Designed reporting mechanism distinguishing legitimate long-lived allocations from leaked memory. Enabled field deployment teams to diagnose memory leaks before service-impacting failures occurred.

**Languages:** C  
**Tools:** GCC, Glibc Hooks, Clearcase, Runtime Analysis  
**Tech:** Mark-Sweep Algorithm, Malloc Interposition, Conservative GC, Graph Traversal

## B-Labs

Nov, 2009 – Sep, 2010

### SR. ENGINEER - CONTRACTOR

London, UK (Remote)

#### CodeZero Hypervisor ARM Port

Nov, 2009 – Sep, 2010

- Synopsis** CodeZero represents microkernel-based hypervisor architecture targeting ARM Cortex-A9 multicore processors. Executed comprehensive platform bring-up enabling bare-metal execution on emerging ARMv7 SMP hardware.
- Details** Ported complete CodeZero hypervisor infrastructure to ARM Cortex-A9 SMP platforms including cache coherency management. CodeZero follows the **L4 microkernel family** model with a minimal kernel focused on address spaces, threads, IPC, and capability-style isolation, which shaped the platform bring-up and guest integration work. Developed **ARM-optimized memcpy** implementation leveraging NEON SIMD instructions for enhanced performance. Extended QEMU device models supporting CodeZero-specific platform requirements. Ported Linux kernel as guest OS executing under CodeZero hypervisor supervision.

**Languages:** C, ARMv7 Assembly  
**Tools:** Git, QEMU, ARM GCC, JTAG Debugger  
**Tech:** ARM Cortex-A9 MPCore, SMP, Cache Coherency, NEON SIMD, Microkernel IPC

## Harman International

Jul. 2009 – Nov. 2009

### ENGINEER

Bengaluru, India

#### QNX BSP for Freescale PowerPC Platforms

Jul. 2009 – Nov. 2009

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Synopsis</b> | Board Support Package development for QNX real-time operating system on Freescale QorIQ P4080 (8-core) and P2020 (dual-core) PowerPC processors targeting automotive infotainment systems. Enabled QNX deployment on next-generation multicore embedded platforms.                                                                                                                                              |
| <b>Details</b>  | Resolved critical <b>SMP initialization bug</b> preventing P4080 8-core bring-up on QNX RTOS. Ported I2C controller driver enabling communication with touchscreen, audio codecs, and sensor peripherals. Developed foundational BSP infrastructure including timer subsystem and interrupt controller drivers. Successfully achieved <b>early boot milestones</b> on both P2020 and P4080 reference platforms. |

**Languages:** C, PowerPC e500 Assembly  
**Tools:** QNX Momentics IDE, Make, PowerPC Cross Compiler  
**Tech:** QorIQ P4080, P2020, e500mc Core, Altivec, SMP Spinlocks, I2C Protocol

## ARM Ltd.

2005 – 2009

### SR. DEVELOPMENT ENGINEER

Bengaluru, India

#### Multi-OS Enablement for ARM Processor Portfolio

Aug. 2005 – Jun. 2009

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Synopsis</b> | Platform engineering team responsible for enabling diverse operating systems on emerging ARM processor cores. Supported major OS ecosystems including Linux, Symbian, and QNX across ARM11 (ARM1176JZF-S with TrustZone) and Cortex-A8 architectures, facilitating early partner evaluation and ecosystem development.                                                                                                                                                     |
| <b>Details</b>  | Developed <b>production-grade resistive touchscreen driver</b> for Symbian OS with multi-touch gesture recognition. Architected <b>Python-based test automation framework</b> reducing Symbian OS validation cycle time by 60%. Designed precision interrupt latency measurement infrastructure quantifying TrustZone secure-world context switch overhead. Ported L4Ka::Pistachio microkernel to ARM architecture enabling virtualization research and proof-of-concepts. |

**Languages:** C, ARMv6/v7 Assembly, Python  
**Tools:** ARM Fast Models, DS-5 Debugger, RealView Compiler, SoCDesigner  
**Tech:** ARM1176JZF-S, Cortex-A8, TrustZone, Symbian OS, L4 Microkernel, VFP

## Sasken Ltd.

2004 – 2005

### SR. SOFTWARE ENGINEER

Bengaluru, India

#### EFFS - Embedded Flash Filesystem for UMTS Infrastructure

Aug. 2004 – Aug. 2005

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Synopsis</b> | Designed and maintained Extended Flash File System (EFFS) for VxWorks-based UMTS base station controllers. Addressed unique constraints of Flash storage including limited erase cycles, asymmetric read/write performance, and power-failure resilience requirements for telecom infrastructure.                                                                                                                                                     |
| <b>Details</b>  | Architected <b>power-failure-resilient filesystem</b> ensuring data integrity across unexpected resets in field deployments. Developed <b>wear-leveling algorithm</b> distributing write operations across Flash blocks extending device lifespan by 3x. Implemented atomic transaction mechanism with flat file tree optimized for Flash characteristics. Integrated filesystem with VxWorks I/O subsystem maintaining POSIX-like API compatibility. |

**Languages:** C  
**Tools:** Tornado IDE, Wind River Compiler, VxWorks RTOS  
**Tech:** NOR Flash, Wear Leveling, Erase Block Management, Power-Fail Safe, POSIX I/O

## Global Edge Software

2003 – 2004

### SOFTWARE ENGINEER

Bengaluru, India

#### Linux SDIO Host Controller Driver for Wireless Connectivity

Jun. 2003 – Aug. 2004

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Synopsis</b> | Developed high-performance SDIO host controller driver for Intel StrongARM embedded platforms enabling wireless connectivity through Marvell 802.11g WiFi chipsets. Optimized for maximum throughput on resource-constrained ARM processors targeting portable and embedded applications.                                                                                                                                                                                           |
| <b>Details</b>  | Architected and implemented complete <b>SDIO host controller driver</b> supporting 4-bit wide bus mode and clock management. Engineered <b>DMA-based transfer mechanism</b> achieving maximum theoretical 802.11g throughput (54 Mbps) on StrongARM platform. Optimized interrupt handling and data transfer pathways minimizing CPU overhead and latency. Developed comprehensive integration with Marvell WiFi chipset firmware achieving production-grade wireless connectivity. |

**Languages:** C  
**Tools:** Git, GCC, GDB, ARM Cross Compiler, Logic Analyzer  
**Tech:** Intel StrongARM SA-1110, SDIO Protocol, 802.11g WiFi, DMA Engine, IRQ Handling



# Awards

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|                                                                                                                                                        |                           |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------|
| Q3 2018                                                                                                                                                | <b>Spotlight</b>          | <i>AMD</i>      |
| Helping Emulation team find design bug in data fabric                                                                                                  |                           |                 |
| Q2 2019                                                                                                                                                | <b>Spotlight</b>          | <i>AMD</i>      |
| Optimization of exp() and improving performance by 30%                                                                                                 |                           |                 |
| Q2 2019                                                                                                                                                | <b>Director Spotlight</b> | <i>AMD</i>      |
| Supporting Arden Ubuntu emulation boot, fixing initramfs delayed mount                                                                                 |                           |                 |
| Q4 2020                                                                                                                                                | <b>Spotlight</b>          | <i>AMD</i>      |
| Providing important patches to GLIBC to fix memcpy behaviour on AMD, open sourcing AMD LibM                                                            |                           |                 |
| Q3 2021                                                                                                                                                | <b>Spotlight</b>          | <i>AMD</i>      |
| Delivering Cryptography PoC (CFB based parallel decrypting), increases performance by 3x                                                               |                           |                 |
| 2016                                                                                                                                                   | <b>Recognition</b>        | <i>Broadcom</i> |
| Upstreaming SMMUv3 emulation and delivering Linux driver infrastructure that enabled early ARMv8 server platform bring-up and validation               |                           |                 |
| 2013                                                                                                                                                   | <b>Recognition</b>        | <i>Cavium</i>   |
| Delivering the MIPS64 Kexec/Kdump port and virtualization bring-up work that improved Octeon III platform readiness and post-mortem debug capability   |                           |                 |
| 2008                                                                                                                                                   | <b>Recognition</b>        | <i>ARM</i>      |
| Driving ARM platform enablement through OS porting, automated validation, and L4Ka::Pistachio microkernel work that expanded early ecosystem readiness |                           |                 |